

A Statistical Learning Framework for Explainable and Transferable Forest Disturbance-Agent Attribution with Provable Guarantees

Khaleel Ahmad^{1*}, Walid Abdelfattah², Malick Sallah^{1,3*}, Jawaher Saud Alrashood⁴,

¹ Department of Mathematics and Statistics, The University of Lahore, 1-KM Defence Road, Lahore 54000, Pakistan; khalil7066616@gmail.com

² Humanities and Social Research Center, Northern Border University, Arar, Saudi Arabia; walidabdefattah@nbu.edu.sa

³ Department of Mathematics, The University of the Gambia, Serrekunda P.O. Box 3530, The Gambia

⁴ Department of Teaching and Learning, College of Education and Human Development, Princess Nourah bint Abdulrahman University, Saudi Arabia; gsalsrshod@pnu.edu.sa

Correspondence: khalil7066616@gmail.com; malick.sallah@utg.edu.gm

Abstract. We develop the mathematical foundations of a framework that attributes a causal agent, namely wildfire, harvest, insect defoliation, windthrow, or drought, to each disturbance event detected in standardized satellite time-series change metrics. The attribution task is posed as a statistical decision problem on a feature space built from spectral-temporal descriptors, object geometry, terrain, and climate. Within this formulation we prove the Bayes optimality of the posterior-maximizing rule and of its cost-sensitive counterpart, a finite-sample generalization bound for the empirical-risk minimizer, the variance-reduction identity that governs ensemble estimators together with a bias-variance decomposition, an ordering between the Gini and entropy impurities, the convexity and second-order optimality conditions of gradient-boosted trees, the gradient identities of the softmax and focal objectives, the efficiency, symmetry, and null-player properties of the Shapley attribution used for explanation, the order preservation of temperature scaling, and the distribution-free lower and upper coverage bounds of the conformal predictor used for uncertainty. We further prove an unbiasedness and variance result for the design-based accuracy estimator and a decomposition of the cross-region transfer gap.

Keywords. disturbance-agent attribution; statistical learning theory; Bayes classifier; Shapley value; conformal prediction; spatial transferability.

MSC 2020. Primary 62H30, 68T05; Secondary 62G05, 62P12, 68T07, 91A12.

1 Introduction

Forest disturbance governs a large share of the variability in the terrestrial carbon balance and in the distribution of habitat, and its accelerating dynamics under a warming climate have made its measurement a central concern of environmental science. Seidl et al. [1] showed that the frequency and severity of forest disturbances are rising across biomes as climate forcing intensifies, while McDowell et al. [2]

documented a pervasive shift toward shorter-lived, more frequently disturbed forests worldwide. The economic and policy stakes of distinguishing the drivers of that change were made concrete by Curtis et al. [3], who classified the dominant drivers of global forest loss and demonstrated that commodity-driven and forestry-driven losses have very different permanence. The observational basis for such studies is the open satellite record, and Hansen et al. [4] established its power at planetary scale by mapping annual forest cover change from Landsat for the entire globe. Woodcock et al. [5] framed the methodological consequence of this data richness as a transition from episodic change detection toward continuous monitoring, and Wulder et al. [6] reviewed the scientific and programmatic status of the Landsat mission that underwrites it. The computational counterpart of this shift is the cloud platform described by Gorelick et al. [7], which relocated planetary-scale analysis from local infrastructure to a shared backend and made large-area time-series processing routine.

The analytical core of disturbance monitoring is temporal segmentation, and several algorithms have shaped the field. Kennedy et al. [8] introduced LandTrendr, which fits idealized trajectories to annual spectral series and reads their vertices as disturbance and recovery segments. Zhu and Woodcock [9] proposed continuous change detection and classification, which models all clear observations with harmonic functions and flags breaks as they occur. Verbesselt et al. [10] developed a decomposition of dense series into trend and seasonal components to isolate structural change. Building on this lineage, Hermosilla et al. [11] defined an integrated Landsat protocol that produces annual gap-free composites suitable for consistent segmentation, and Hermosilla et al. [12] extended it into a continental system that detects, characterizes, and attributes annual forest change from time-series metrics. Hermosilla et al. [13] then described the mass data-processing pipeline that turns raw pixels into these data products at national scale, and White et al. [14] applied the resulting metrics to characterize twenty-five years of disturbance and recovery across an entire country. Most recently, Hermosilla et al. [15] reimplemented this protocol, Composite2Change, as a native cloud function, so that its standardized change metrics, namely magnitude, duration, rate, and recovery, are now reproducible across sensors and regions without local infrastructure.

That change metrics computed by different algorithms need not agree was quantified by Cohen et al. [16], who built a multispectral ensemble of LandTrendr fits, and by Cohen et al. [17], who showed that diversity of algorithm and spectral input improves the monitoring of disturbance. These findings motivate working on a single standardized metric product rather than on any one algorithm's idiosyncratic output. The step from detecting change to attributing its cause has its own literature. Schroeder et al. [18] separated wildfire from clearcut harvest in boreal forests using trajectory features and simple decision rules. Kennedy et al. [19] attributed change agents from LandTrendr trajectories in support of habitat monitoring, establishing supervised attribution as a workable approach. Oeser et al. [20] exploited intra-annual Landsat observations to attribute disturbance agents in a managed temperate landscape, and Sebald et al. [21] demonstrated that landscape context substantially improves the separation of human from natural disturbance. Stahl et al. [22] advanced an automated attribution of disturbance types combining time-series features with ancillary data. On the ecological side, Senf et al. [23] showed that canopy mortality in temperate Europe has roughly doubled over recent decades, and Senf and Seidl [24] mapped the continent's disturbance regimes, both of which underline the need for agent-resolved information at scale. The complementary reason attribution matters is dynamical: Coops et al. [25] showed that non-stand-replacing disturbances leave a distinct signature in condition trajectories, and Frazier et al. [26] showed that post-fire recovery rates vary systematically in space and time, so the agent controls the trajectory that follows.

The learning machinery on which attribution rests is well developed. Breiman [27] introduced random

forests, whose accuracy rests on decorrelating an ensemble of trees. Friedman [28] formalized gradient boosting as stagewise additive modelling, and Chen and Guestrin [29] gave the regularized, second-order variant now standard in practice. For sequential inputs, Pelletier et al. [30] adapted temporal convolutional networks to satellite image time series, and Vaswani et al. [31] introduced the self-attention architecture that has since become dominant for sequence modelling. Interpretation and reliability have become inseparable from deployment. Shapley [32] defined the unique fair allocation of value in a cooperative game, and Lundberg and Lee [33] turned that construction into a unified, additive method for explaining model predictions. Guo et al. [34] showed that modern classifiers are frequently miscalibrated and that simple post-hoc scaling restores reliability, while Angelopoulos and Bates [35] gave an accessible account of conformal prediction, which yields prediction sets with a finite-sample coverage guarantee. Credible evaluation over space imposes its own discipline: Roberts et al. [36] catalogued cross-validation strategies for spatially and temporally structured data, and Ploton et al. [37] demonstrated that ignoring spatial autocorrelation produces badly optimistic accuracy estimates. The design-based estimation of map accuracy and area was placed on a rigorous footing by Olofsson et al. [38], whose good-practice estimators we adopt, and the class imbalance endemic to disturbance data is addressed by the focal objective of Lin et al. [39].

Against this background the gap is specific. The standardized metrics of Hermosilla et al. [15] have made the questions of *when* a disturbance occurred and *how* it unfolded reproducible at global scale, yet the causal question of *which* agent was responsible is still answered by pipelines that are tied to a particular segmentation algorithm and a particular region, that report a single hard label without a statement of confidence, and that rarely expose the evidence behind a decision. The attribution studies cited above, from Schroeder et al. [18] to Stahl et al. [22], are valuable but share these properties, so their models transfer poorly and their outputs are difficult to certify for accounting. What has been missing is a treatment in which attribution is defined on the standardized metric product itself, is equipped with explanation and with distribution-free uncertainty, and, above all, rests on results whose correctness is established by proof rather than by empirical demonstration alone. The present paper supplies that treatment. We formalize disturbance-agent attribution as a decision problem on an object-level feature space, and we prove the properties that make the framework well posed: the optimality of the decision rule and of its cost-sensitive form, a finite-sample generalization bound, the variance behaviour and bias-variance decomposition that justify ensembling, an ordering of impurity measures, the convexity and optimality conditions of the boosting step, the gradient identities of the training objectives, the axiomatic properties of the Shapley explanation, the order preservation of calibration, the coverage of the conformal predictor, the unbiasedness and variance of the accuracy estimator, and a decomposition of the transfer gap. Section 2 fixes notation and records the definitions we rely on. Section 3 states and proves the main results and illustrates them with worked examples. Section 4 compares the framework with existing attribution approaches, Section 5 presents tabulated numerical illustrations, and Section 6 concludes.

2 Preliminaries

Throughout, feature vectors live in a Borel subset of a Euclidean space and labels in a finite set. We collect the definitions used later, and fix notation. We write K for the number of disturbance-agent classes and M for the number of features describing a change object, and we abbreviate $\eta_k(\mathbf{x}) = \mathbb{P}(y = k \mid \mathbf{x})$ for the posterior probabilities.

Definition 1: [40] A measurable feature map assigns to each change object a vector $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^M$, and its disturbance agent is a label $y \in \mathcal{Y} = \{1, \dots, K\}$. The pair $(\mathbf{x}, y)$ is a random element with joint distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$, factorized as $\mathcal{D}(d\mathbf{x}, k) = p(\mathbf{x}) \eta_k(\mathbf{x}) d\mathbf{x}$. A classifier is a measurable map $h : \mathcal{X} \rightarrow \mathcal{Y}$, and a scoring rule is a measurable map $\hat{\eta} : \mathcal{X} \rightarrow \Delta^{K-1}$ into the probability simplex.

Definition 2: [40] For a loss function $L : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}$, the risk of a classifier h is $R(h) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}}[L(y, h(\mathbf{x}))]$. The zero-one loss is $L(k, l) = \mathbb{I}[k \neq l]$, and a cost-sensitive loss is given by a nonnegative matrix $C = (C_{kl})$ with $C_{kk} = 0$, where C_{kl} is the cost of predicting l when the truth is k .

Definition 3: [41] Given a hypothesis class $\mathcal{H}$ of classifiers and a sample $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ drawn independently from $\mathcal{D}$, the empirical risk of $h \in \mathcal{H}$ is $R_n(h) = \frac{1}{n} \sum_{i=1}^n L(y_i, h(\mathbf{x}_i))$, and the empirical-risk minimizer is $\hat{h} \in \arg \min_{h \in \mathcal{H}} R_n(h)$.

Definition 4: [40] The Bayes classifier for the zero-one loss is any classifier returning a most probable class, $h^*(\mathbf{x}) \in \arg \max_{k \in \mathcal{Y}} \eta_k(\mathbf{x})$, and the Bayes risk is $R^* = R(h^*)$.

Definition 5: [42] For a node t carrying training objects with class proportions $\hat{p}_{t,k}$, the Gini impurity is $G(t) = \sum_{k=1}^K \hat{p}_{t,k}(1 - \hat{p}_{t,k}) = 1 - \sum_{k=1}^K \hat{p}_{t,k}^2$, and the entropy is $H(t) = -\sum_{k=1}^K \hat{p}_{t,k} \log \hat{p}_{t,k}$. A split sending fractions π_L and $\pi_R = 1 - \pi_L$ of the objects to two children is scored by the impurity decrease $\Delta = G(t) - \pi_L G(t_L) - \pi_R G(t_R)$.

Definition 6: [27] A random forest is an average of B classification trees, each grown on an independent bootstrap resample of the training set and restricted at every split to a random subset of features. Writing $\hat{\eta}_b(\mathbf{x})$ for the class-probability vector of tree b , the forest estimate is $\bar{\eta}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \hat{\eta}_b(\mathbf{x})$.

Definition 7: [28] A gradient-boosted tree model is an additive expansion $F(\mathbf{x}) = \sum_{m=1}^{T_{\text{ens}}} f_m(\mathbf{x})$ in which each f_m is a regression tree fitted stagewise. With a differentiable loss $\ell(y, \hat{y})$ and a tree penalty $\Omega(f) = \gamma T + \frac{1}{2} \lambda \|\mathbf{w}\|^2$ on a tree with T leaves and weights $\mathbf{w}$, the round- t objective is $\mathcal{L}^{(t)} = \sum_{i=1}^N \ell(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t)$.

Definition 8: [43] The softmax map sends a score vector $\mathbf{z} \in \mathbb{R}^K$ to the probability vector with components $p_k = e^{z_k} / \sum_{j=1}^K e^{z_j}$. The cross-entropy loss against a one-hot label $\mathbf{y}$ is $\ell(\mathbf{y}, \mathbf{z}) = -\sum_{k=1}^K y_k \log p_k$.

Definition 9: [39] For class weights $\alpha_k > 0$ and a focusing parameter $\gamma \geq 0$, the focal loss against a one-hot label is $\mathcal{L}_{\text{focal}} = -\sum_{k=1}^K \alpha_k (1 - p_k)^\gamma y_k \log p_k$, where p_k is the softmax probability of Definition 8.

Definition 10: [31] Given a sequence embedded as rows of $X \in \mathbb{R}^{n \times d_{\text{model}}}$ and projections $Q = XW_Q$, $K = XW_K$, $V = XW_V$ with keys of dimension d_k , scaled dot-product attention returns $\text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k}) V$.

Definition 11: [32] Let $\mathcal{M} = \{1, \dots, M\}$ be a set of players and $v : 2^{\mathcal{M}} \rightarrow \mathbb{R}$ a value function with $v(\emptyset) = 0$. The Shapley value of player i is $\phi_i = \sum_{S \subseteq \mathcal{M} \setminus \{i\}} \frac{|S|!(M-|S|-1)!}{M!} [v(S \cup \{i\}) - v(S)]$.

Definition 12: [33] For a model f and an input $\mathbf{x}$, an additive feature attribution defines the value function $v(S) = \mathbb{E}[f(\mathbf{x}) \mid \mathbf{x}_S]$ on feature coalitions $S \subseteq \mathcal{M}$ and explains the prediction by $g(\mathbf{x}) = \phi_0 + \sum_{i=1}^M \phi_i$, where $\phi_0 = \mathbb{E}[f(\mathbf{x})]$ and ϕ_i are the Shapley values of Definition 11 for this v .

Definition 13: [34] A scoring rule is calibrated if $\mathbb{P}(y = \hat{y} \mid \hat{p} = q) = q$ for all confidences q , where $\hat{p} = \max_k p_k$ is the reported confidence and $\hat{y}$ the predicted class. Its expected calibration error is $\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$, the average gap between accuracy and confidence over B confidence bins. Temperature scaling replaces $\mathbf{z}$ by $\mathbf{z}/T$ for a single scalar $T > 0$ chosen on held-out data.

Definition 14: [44] Random elements $Z_1, \dots, Z_n$ are exchangeable if their joint law is invariant under every permutation. Given a nonconformity score $s(\mathbf{x}, y)$ and a calibration set of exchangeable labelled pairs, the split-conformal predictor at level $\alpha \in (0, 1)$ returns $\mathcal{C}(\mathbf{x}) = \{y \in \mathcal{Y} : s(\mathbf{x}, y) \leq \hat{q}\}$, where $\hat{q}$ is the $\lceil (n+1)(1-\alpha) \rceil$ smallest calibration score.

Definition 15: [36] A spatially blocked K -fold partition assigns whole contiguous spatial blocks, rather than individual objects, to folds, so that objects in the held-out fold are geographically separated from those used for training. The cross-validation risk estimator is $\hat{R}_{CV} = \frac{1}{K} \sum_{k=1}^K \frac{1}{|\mathcal{D}_k|} \sum_{i \in \mathcal{D}_k} L(y_i, \hat{h}_{-k}(\mathbf{x}_i))$, where $\hat{h}_{-k}$ is trained on all folds but the k -th.

Figure 1 summarizes how these objects compose into the attribution framework whose properties Section 3 establishes.

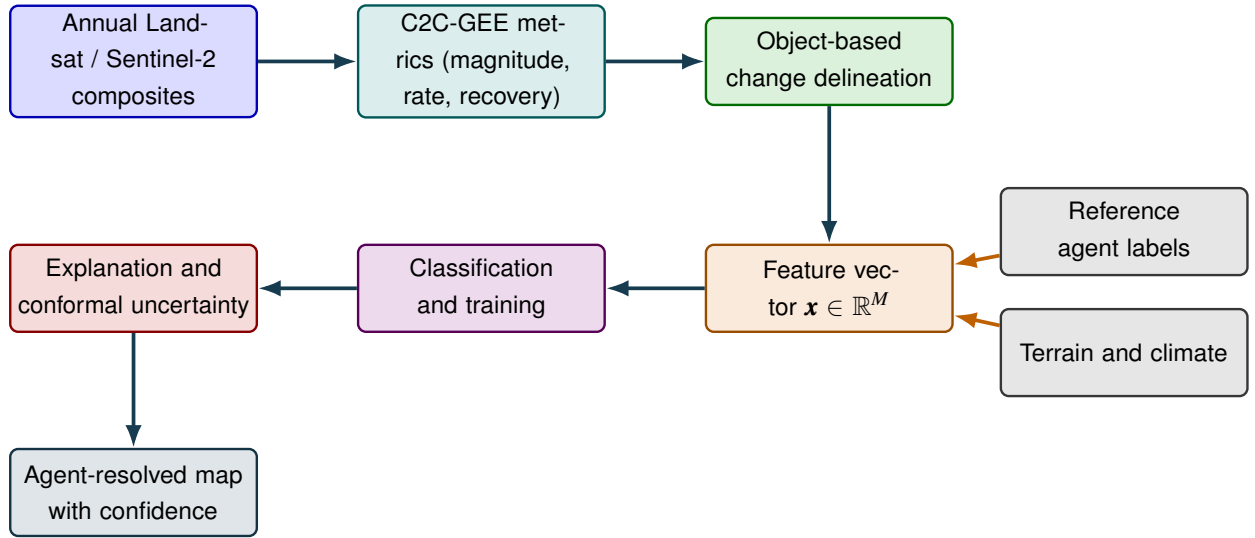


Figure 1. Composition of the attribution framework. Standardized change metrics (Definition 1) are aggregated into change objects, joined with reference labels and ancillary covariates to form the feature vector, classified by the estimators of Definitions 6, 7 and 10, and returned with the Shapley explanation of Definition 12 and the conformal set of Definition 14.

3 Main results

We establish the properties of the framework, ordered from the decision-theoretic foundation through the statistical guarantees, the estimators used for classification, explanation, calibration, and uncertainty, to the evaluation of accuracy and transfer. Each result is stated for the attribution setting of Section 2 and proved in full, and several are accompanied by a worked numerical example.

The first result identifies the classifier that any training procedure ought to approximate.

Theorem 1 (Bayes optimality). *Under the zero-one loss of Definition 2, every classifier h satisfies $R(h) \geq R^*$, where $R^* = R(h^*)$ is attained by the Bayes classifier h^* of Definition 4.*

Proof. For the zero-one loss the risk equals the probability of error, which we condition on the features:

$$R(h) = \mathbb{P}(h(\mathbf{x}) \neq y) = \mathbb{E}_{\mathbf{x}}[\mathbb{P}(y \neq h(\mathbf{x}) \mid \mathbf{x})] = \mathbb{E}_{\mathbf{x}}[1 - \eta_{h(\mathbf{x})}(\mathbf{x})]. \quad (1)$$

For each fixed $\mathbf{x}$ the integrand is minimized by the label with the largest posterior, since for any value $h(\mathbf{x}) = l$ we have $\eta_l(\mathbf{x}) \leq \max_k \eta_k(\mathbf{x}) = \eta_{h^*(\mathbf{x})}(\mathbf{x})$, hence $1 - \eta_{h(\mathbf{x})}(\mathbf{x}) \geq 1 - \eta_{h^*(\mathbf{x})}(\mathbf{x})$. Taking expectations preserves the inequality, so $R(h) \geq \mathbb{E}_{\mathbf{x}}[1 - \eta_{h^*(\mathbf{x})}(\mathbf{x})] = R^*$, with equality attained by h^* . $\square$

When the two kinds of error carry different costs, the optimal rule changes in a way that justifies the class weighting used in training.

Proposition 1 (Cost-sensitive optimal rule). *Under the cost-sensitive loss $L(k, l) = C_{kl}$ of Definition 2, the risk is minimized by $h_C^*(\mathbf{x}) = \arg \min_{l \in \mathcal{Y}} \sum_{k=1}^K C_{kl} \eta_k(\mathbf{x})$.*

Proof. Conditioning as in Theorem 1, $R(h) = \mathbb{E}_{\mathbf{x}}[\sum_k C_{k, h(\mathbf{x})} \eta_k(\mathbf{x})]$. The inner sum depends on h only through the predicted label $l = h(\mathbf{x})$ and is minimized pointwise by the label with the smallest conditional expected cost $\sum_k C_{kl} \eta_k(\mathbf{x})$. Selecting it for every $\mathbf{x}$ defines h_C^* and minimizes the expectation. With $C_{kl} = \mathbb{I}[k \neq l]$ the conditional cost is $1 - \eta_l(\mathbf{x})$ and the rule reduces to Theorem 1, which is therefore a special case. $\square$

The rule of Theorem 1 is unavailable in practice because η is unknown, so a classifier is chosen from a hypothesis class by minimizing empirical risk. Its risk splits into two transparent pieces.

Proposition 2 (Excess-risk decomposition). *For any $\hat{h} \in \mathcal{H}$, writing $R_{\mathcal{H}} = \inf_{h \in \mathcal{H}} R(h)$,*

$$R(\hat{h}) - R^* = \underbrace{(R(\hat{h}) - R_{\mathcal{H}})}_{\text{estimation error}} + \underbrace{(R_{\mathcal{H}} - R^*)}_{\text{approximation error}}. \quad (2)$$

Moreover the estimation error is bounded by twice the uniform deviation, $R(\hat{h}) - R_{\mathcal{H}} \leq 2 \sup_{h \in \mathcal{H}} |R(h) - R_n(h)|$ whenever $\hat{h}$ minimizes R_n over $\mathcal{H}$.

Proof. Identity (2) follows by adding and subtracting $R_{\mathcal{H}}$. For the bound, let $h^\dagger$ attain $R_{\mathcal{H}}$ up to arbitrarily small $\varepsilon > 0$. Since $\hat{h}$ minimizes empirical risk, $R_n(\hat{h}) \leq R_n(h^\dagger)$, and therefore

$$R(\hat{h}) - R_{\mathcal{H}} \leq (R(\hat{h}) - R_n(\hat{h})) + (R_n(h^\dagger) - R(h^\dagger)) + \varepsilon \leq 2 \sup_{h \in \mathcal{H}} |R(h) - R_n(h)| + \varepsilon. \quad (3)$$

Letting $\varepsilon \downarrow 0$ gives the claim. $\square$

The uniform deviation appearing in Proposition 2 is controlled by a concentration inequality applied uniformly over a finite class. We use the following classical bound.

Lemma 1 (Hoeffding bound). *[45] If $X_1, \dots, X_n$ are independent with $X_i \in [0, 1]$ and $\bar{X} = \frac{1}{n} \sum_i X_i$, then for every $t > 0$, $\mathbb{P}(\bar{X} - \mathbb{E}\bar{X} \geq t) \leq e^{-2nt^2}$ and $\mathbb{P}(|\bar{X} - \mathbb{E}\bar{X}| \geq t) \leq 2e^{-2nt^2}$.*

Theorem 2 (Finite-class generalization bound). *Let $\mathcal{H}$ be finite with $|\mathcal{H}| = m$ and let the loss take values in $[0, 1]$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the sample, every $h \in \mathcal{H}$ satisfies*

$$R(h) \leq R_n(h) + \sqrt{\frac{\log m + \log(1/\delta)}{2n}}. \quad (4)$$

Consequently the empirical-risk minimizer $\hat{h}$ obeys $R(\hat{h}) - R_{\mathcal{H}} \leq 2\sqrt{(\log m + \log(1/\delta))/(2n)}$.

Proof. Fix $h \in \mathcal{H}$. The losses $L(y_i, h(\mathbf{x}_i)) \in [0, 1]$ are independent with mean $R(h)$, and their average is $R_n(h)$, so Lemma 1 gives $\mathbb{P}(R(h) - R_n(h) \geq t) \leq e^{-2nt^2}$. Applying the union bound over the m hypotheses,

$$\mathbb{P}\left(\exists h \in \mathcal{H} : R(h) - R_n(h) \geq t\right) \leq m e^{-2nt^2}. \quad (5)$$

Setting the right-hand side equal to δ and solving gives $t = \sqrt{(\log m + \log(1/\delta))/(2n)}$, which yields (4) on the complementary event of probability at least $1 - \delta$. The bound on the estimation error follows by inserting (4) into the inequality of Proposition 2 and using $R_n(\hat{h}) \leq R_n(h^\dagger)$ once more. $\square$

Example 1. Take a class of $m = 1000$ candidate attribution rules, a sample of $n = 5000$ labelled change objects, and $\delta = 0.05$. Then $\log m + \log(1/\delta) = \log 1000 + \log 20 = 6.9078 + 2.9957 = 9.9035$, so the uniform bound in (4) is $\sqrt{9.9035/10000} = 0.0315$. With probability at least 0.95, the true risk of every rule in the class exceeds its empirical risk by at most about three percentage points, and the empirical minimizer is within about six points of the best achievable risk in the class.

The two impurity measures of Definition 5 used to grow trees are ordered, which clarifies why either can be used interchangeably as a splitting criterion.

Lemma 2 (Gini-entropy inequality). [46] *For any probability vector $\mathbf{p} = (p_1, \dots, p_K)$, the Gini impurity and the entropy in nats satisfy $G(\mathbf{p}) \leq H(\mathbf{p})$, with equality if and only if some $p_k = 1$.*

Proof. The elementary inequality $\log x \leq x - 1$ for $x > 0$, applied at $x = p_k$ for each k with $p_k > 0$, gives $\log p_k \leq p_k - 1$, hence $-\log p_k \geq 1 - p_k$. Multiplying by $p_k \geq 0$ yields $-p_k \log p_k \geq p_k(1 - p_k)$. Summing over k gives $H(\mathbf{p}) = \sum_k (-p_k \log p_k) \geq \sum_k p_k(1 - p_k) = G(\mathbf{p})$. Equality in $\log x \leq x - 1$ holds only at $x = 1$, so the summed inequality is tight exactly when every $p_k \in \{0, 1\}$, that is when one coordinate equals 1. $\square$

Figure 2 displays the two impurities in the binary case, where the ordering of Lemma 2 is visible.

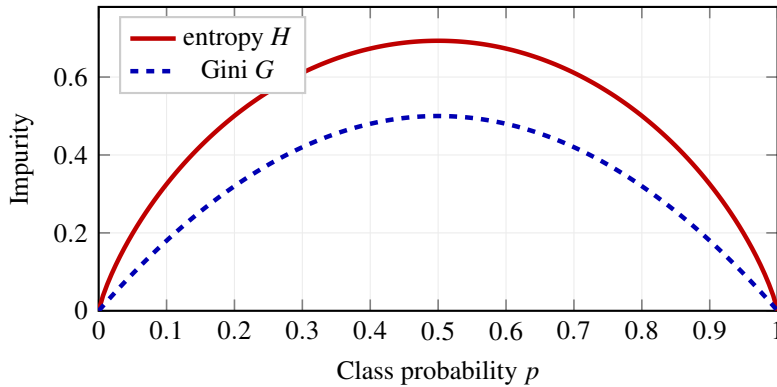


Figure 2. Binary entropy and Gini impurity against the class probability. The entropy dominates the Gini index everywhere, in agreement with Lemma 2, and the two agree only at the degenerate distributions.

The forest of Definition 6 estimates the posterior by averaging trees. The following identity explains why averaging helps and why decorrelation, rather than the number of trees alone, controls the residual variance.

Theorem 3 (Ensemble variance reduction). *Fix a class and let $\hat{\eta}_1, \dots, \hat{\eta}_B$ be identically distributed tree estimators of its posterior probability, each with variance σ^2 and common pairwise correlation ρ . Then*

the forest estimator $\bar{\eta} = \frac{1}{B} \sum_{b=1}^B \hat{\eta}_b$ has

$$\text{Var}(\bar{\eta}) = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2, \quad (6)$$

and $\text{Var}(\bar{\eta}) \downarrow \rho \sigma^2$ as $B \rightarrow \infty$.

Proof. By bilinearity of covariance,

$$\text{Var}(\bar{\eta}) = \frac{1}{B^2} \sum_{b=1}^B \sum_{b'=1}^B \text{Cov}(\hat{\eta}_b, \hat{\eta}_{b'}) = \frac{1}{B^2} [B\sigma^2 + B(B-1)\rho\sigma^2], \quad (7)$$

since the B diagonal terms each equal σ^2 and the $B(B-1)$ off-diagonal terms each equal $\rho\sigma^2$. Simplifying gives (6). The second term is nonnegative and strictly decreasing in B when $\rho < 1$, so the variance decreases monotonically to $\rho\sigma^2$, the floor set by the inter-tree correlation that bootstrap resampling and random feature selection act to reduce. $\square$

Figure 3 plots the identity and makes the role of ρ visible.

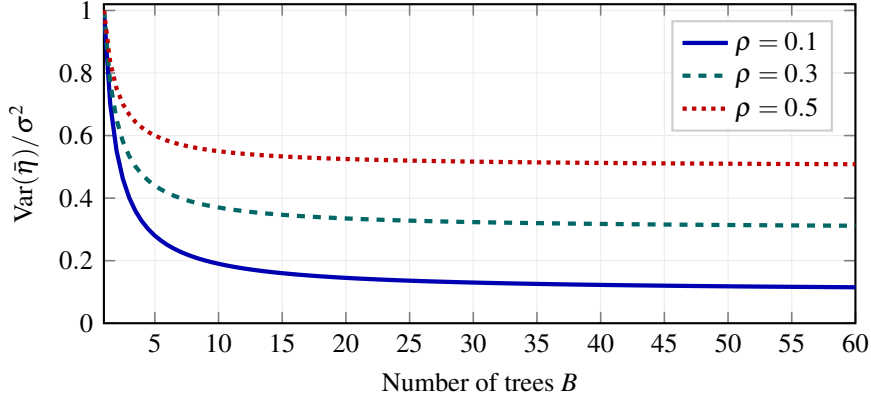


Figure 3. Normalized forest variance from Theorem 3 against ensemble size for three inter-tree correlations. Each curve decreases to its floor ρ , so lowering ρ , not merely increasing B , reduces variance.

The same averaging argument governs the object-level features of the framework, whose per-object summaries average many correlated pixels.

Corollary 1 (Object-aggregation stabilization). *If a change object aggregates n_o pixel measurements of a feature, each with variance τ^2 and common pairwise correlation ψ , then the object mean has variance $\psi\tau^2 + \frac{1-\psi}{n_o}\tau^2$.*

Proof. The pixel measurements play the role of the trees in Theorem 3: they are identically distributed with variance τ^2 and common correlation ψ , and their average is the object-level feature. Substituting τ, ψ, n_o for σ, ρ, B in (6) gives the stated variance. Thus aggregation reduces the per-object measurement noise toward the floor $\psi\tau^2$, which is why object-level features are more stable than single-pixel values. $\square$

For a scoring estimator the quality of the posterior estimate decomposes into interpretable parts.

Proposition 3 (Bias-variance decomposition). *Let $\hat{\eta}_k(\mathbf{x})$ estimate $\eta_k(\mathbf{x})$ at a fixed $\mathbf{x}$ over random training samples. Then*

$$\mathbb{E}[(\hat{\eta}_k(\mathbf{x}) - \eta_k(\mathbf{x}))^2] = (\mathbb{E}[\hat{\eta}_k(\mathbf{x})] - \eta_k(\mathbf{x}))^2 + \text{Var}(\hat{\eta}_k(\mathbf{x})). \quad (8)$$

Proof. Write $\mu = \mathbb{E}[\hat{\eta}_k(\mathbf{x})]$ and insert it: $\hat{\eta}_k - \eta_k = (\hat{\eta}_k - \mu) + (\mu - \eta_k)$. Squaring and taking expectations, $\mathbb{E}[(\hat{\eta}_k - \eta_k)^2] = \mathbb{E}[(\hat{\eta}_k - \mu)^2] + 2(\mu - \eta_k)\mathbb{E}[\hat{\eta}_k - \mu] + (\mu - \eta_k)^2$. The cross term vanishes because $\mathbb{E}[\hat{\eta}_k - \mu] = 0$, leaving $\text{Var}(\hat{\eta}_k) + (\mu - \eta_k)^2$, which is (8). Theorem 3 reduces the variance term, while the bias term reflects the approximation capacity of the estimator. $\square$

Example 2. With inter-tree correlation $\rho = 0.3$ and unit per-tree variance, Theorem 3 gives normalized forest variance $0.3 + 0.7/B$. For $B = 10$ this is 0.37, for $B = 100$ it is 0.307, and the limiting value is 0.30. Almost all of the reducible variance is removed by the first hundred trees, and no number of trees can reduce the variance below the correlation floor 0.30, which illustrates the mechanism of Theorem 3.

The boosting step of Definition 7 relies on the curvature of the loss, so we first record that the cross-entropy is convex in the scores.

Lemma 3 (Convexity of cross-entropy). *The cross-entropy of Definition 8 has Hessian $\nabla_{\mathbf{z}}^2 \ell = \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$, which is positive semidefinite. Hence ℓ is convex in $\mathbf{z}$, and the diagonal curvatures $h_i = p_i(1 - p_i)$ are nonnegative.*

Proof. Differentiating the gradient $\partial \ell / \partial z_j = p_j - y_j$ established in Theorem 5 below gives $\partial^2 \ell / \partial z_j \partial z_k = \partial p_j / \partial z_k = p_j(\delta_{jk} - p_k)$ by Lemma 4, so the Hessian is $\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$. For any $\mathbf{u} \in \mathbb{R}^K$,

$$\mathbf{u}^\top (\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top) \mathbf{u} = \sum_k p_k u_k^2 - \left(\sum_k p_k u_k \right)^2 = \text{Var}_{\mathbf{p}}(U) \geq 0, \quad (9)$$

where U is the random variable taking value u_k with probability p_k ; the expression is its variance and is therefore nonnegative. Thus the Hessian is positive semidefinite and ℓ is convex, and the diagonal entries $p_j(1 - p_j)$ are nonnegative. $\square$

The boosting objective now admits a closed-form optimum, and Lemma 3 guarantees the positivity used in its statement.

Theorem 4 (Optimal leaves and split gain). *Let $g_i = \partial_{\hat{y}} \ell|_{\hat{y}_i^{(t-1)}}$ and $h_i = \partial_{\hat{y}}^2 \ell|_{\hat{y}_i^{(t-1)}} \geq 0$, and for a fixed tree structure put $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$ over the instance set I_j of leaf j . If $\lambda > 0$, then $H_j + \lambda > 0$ and the second-order approximation of the objective in Definition 7 is minimized by*

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \quad \widetilde{\mathcal{L}}^* = -\frac{1}{2} \sum_j \frac{G_j^2}{H_j + \lambda} + \gamma T, \quad (10)$$

and a split of one leaf into children L, R decreases the objective by

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma. \quad (11)$$

Proof. Expanding ℓ to second order about $\hat{y}_i^{(t-1)}$ and discarding the constant term, the objective is $\widetilde{\mathcal{L}}^{(t)} = \sum_i [g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t(\mathbf{x}_i)^2] + \gamma T + \frac{1}{2} \lambda \sum_j w_j^2$. A tree is constant on each leaf, so grouping by leaf gives $\sum_j [G_j w_j + \frac{1}{2} (H_j + \lambda) w_j^2] + \gamma T$. By Lemma 3 each $h_i \geq 0$, so $H_j \geq 0$ and $H_j + \lambda > 0$ for $\lambda > 0$; each summand is then a strictly convex quadratic in w_j whose unique minimizer solves $G_j + (H_j + \lambda) w_j = 0$, giving $w_j^* = -G_j / (H_j + \lambda)$. Substituting back yields $-\frac{1}{2} G_j^2 / (H_j + \lambda)$ per leaf and hence $\widetilde{\mathcal{L}}^*$. Applying (10) to a parent leaf and to its two children and subtracting gives (11), in which the additional leaf contributes $-\gamma$, so a split is accepted only when its gain is positive. $\square$

Example 3. Consider a leaf with aggregated statistics $G_L = -4$, $H_L = 3$ on the left and $G_R = 6$, $H_R = 5$ on the right, with $\lambda = 1$ and $\gamma = 0.5$. Then (11) gives $\text{Gain} = \frac{1}{2}[16/4 + 36/6 - 4/9] - 0.5 = \frac{1}{2}[4 + 6 - 0.444] - 0.5 = 4.778 - 0.5 = 4.278 > 0$, so the split is accepted, and the optimal child weights are $w_L^* = -(-4)/4 = 1.0$ and $w_R^* = -6/6 = -1.0$ by (10).

The neural members of the framework and the boosted logits are trained through the softmax of Definition 8, whose Jacobian and gradient we now record.

Lemma 4 (Softmax Jacobian). *For the softmax of Definition 8, $\frac{\partial p_k}{\partial z_j} = p_k(\delta_{kj} - p_j)$.*

Proof. Write $S = \sum_r e^{z_r}$. For $j = k$ the quotient rule gives $\partial p_k / \partial z_k = (e^{z_k} S - e^{2z_k}) / S^2 = p_k - p_k^2 = p_k(1 - p_k)$, and for $j \neq k$, $\partial p_k / \partial z_j = -e^{z_k} e^{z_j} / S^2 = -p_k p_j$. The two cases combine as $p_k(\delta_{kj} - p_j)$. $\square$

Theorem 5 (Cross-entropy gradient). *For the cross-entropy of Definition 8 against a one-hot label $\mathbf{y}$, $\frac{\partial \ell}{\partial z_j} = p_j - y_j$.*

Proof. By the chain rule and Lemma 4, $\partial \ell / \partial z_j = -\sum_k (y_k / p_k) p_k(\delta_{kj} - p_j) = -\sum_k y_k(\delta_{kj} - p_j)$. Splitting the sum, $-\sum_k y_k \delta_{kj} = -y_j$ and $\sum_k y_k p_j = p_j \sum_k y_k = p_j$ since $\sum_k y_k = 1$, so the derivative equals $p_j - y_j$. $\square$

Temperature scaling and the two extreme limits of the softmax clarify how confidence responds to the temperature.

Proposition 4 (Softmax temperature limits). *For fixed logits $\mathbf{z}$ with a unique maximal coordinate $m = \arg \max_k z_k$, the temperature-scaled softmax satisfies $\lim_{T \rightarrow \infty} p_k(T) = 1/K$ for every k and $\lim_{T \rightarrow 0^+} p_k(T) = \mathbb{I}[k = m]$.*

Proof. Write $p_k(T) = 1 / \sum_j e^{(z_j - z_k)/T}$. As $T \rightarrow \infty$ each exponent $(z_j - z_k)/T \rightarrow 0$, so every summand tends to 1, the denominator tends to K , and $p_k(T) \rightarrow 1/K$. As $T \rightarrow 0^+$, consider $k = m$: for $j \neq m$ we have $z_j - z_m < 0$, so $(z_j - z_m)/T \rightarrow -\infty$ and those summands vanish, leaving the denominator tending to 1, hence $p_m(T) \rightarrow 1$. For $k \neq m$ the summand with $j = m$ has $z_m - z_k > 0$, so $(z_m - z_k)/T \rightarrow +\infty$, the denominator diverges, and $p_k(T) \rightarrow 0$. This is $\mathbb{I}[k = m]$. $\square$

The focal loss of Definition 9 modifies the gradient of Theorem 5 so as to concentrate learning on the rare disturbance agents, and it degenerates to weighted cross-entropy when the focusing parameter vanishes.

Proposition 5 (Focal gradient and its limit). *For a one-hot label with true class c , the focal loss of Definition 9 has gradient*

$$\frac{\partial \mathcal{L}_{\text{focal}}}{\partial z_j} = \alpha_c (1 - p_c)^{\gamma-1} [\gamma p_c \log p_c - (1 - p_c)] (\delta_{cj} - p_j), \quad (12)$$

and setting $\gamma = 0$ recovers the weighted cross-entropy gradient $\alpha_c(p_j - y_j)$.

Proof. With a one-hot label, $\mathcal{L}_{\text{focal}} = -\alpha_c (1 - p_c)^\gamma \log p_c$, a function of p_c alone. Differentiating the product, $\partial \mathcal{L}_{\text{focal}} / \partial p_c = -\alpha_c [-\gamma (1 - p_c)^{\gamma-1} \log p_c + (1 - p_c)^\gamma / p_c]$. By Lemma 4, $\partial p_c / \partial z_j = p_c(\delta_{cj} -$

p_j), so the chain rule gives

$$\begin{aligned}\frac{\partial \mathcal{L}_{\text{focal}}}{\partial z_j} &= \alpha_c [\gamma(1-p_c)^{\gamma-1} \log p_c - (1-p_c)^\gamma / p_c] p_c (\delta_{cj} - p_j) \\ &= \alpha_c (1-p_c)^{\gamma-1} [\gamma p_c \log p_c - (1-p_c)] (\delta_{cj} - p_j),\end{aligned}\quad (13)$$

which is (12). When $\gamma = 0$ the bracket becomes $-(1-p_c)$ and cancels the prefactor $(1-p_c)^{-1}$, leaving $-\alpha_c(\delta_{cj} - p_j) = \alpha_c(p_j - y_j)$. For $\gamma > 0$ the factor $(1-p_c)^\gamma$ in (13) suppresses the gradient of confident objects with $p_c \rightarrow 1$, the intended focusing effect. $\square$

Figure 4 shows how the modulating factor reshapes the contribution of each object.

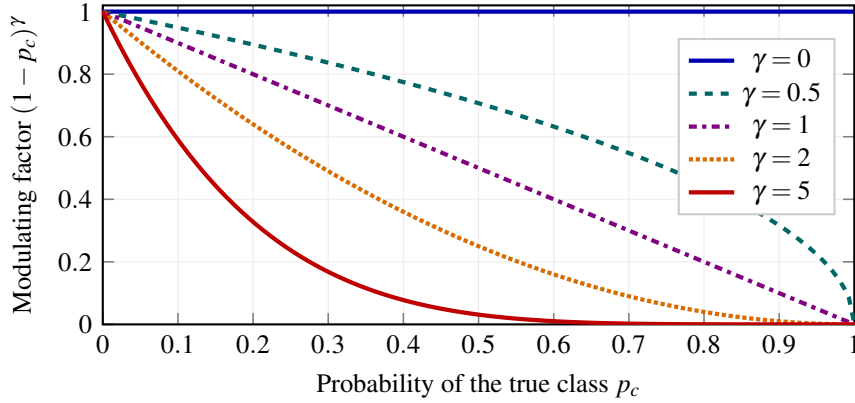


Figure 4. Focal modulating factor of Proposition 5 against the confidence in the true class. Larger γ down-weights well-classified objects more aggressively, so training concentrates on the hard, minority disturbance agents.

The attention mechanism of Definition 10 carries a scaling factor whose necessity is a variance statement.

Proposition 6 (Attention scaling). *If the entries of a query row $\mathbf{q}$ and a key row $\mathbf{k} \in \mathbb{R}^{d_k}$ are independent with zero mean and unit variance, then $\mathbb{E}[\mathbf{q} \cdot \mathbf{k}] = 0$ and $\text{Var}(\mathbf{q} \cdot \mathbf{k}) = d_k$, so dividing the dot product by $\sqrt{d_k}$ produces a score of unit variance.*

Proof. Write $\mathbf{q} \cdot \mathbf{k} = \sum_{i=1}^{d_k} q_i k_i$. By independence and zero means $\mathbb{E}[q_i k_i] = 0$, so the sum has mean zero. The products $q_i k_i$ are independent and each has variance $\mathbb{E}[q_i^2 k_i^2] - 0 = \mathbb{E}[q_i^2] \mathbb{E}[k_i^2] = 1$, so $\text{Var}(\mathbf{q} \cdot \mathbf{k}) = \sum_i 1 = d_k$. Dividing by $\sqrt{d_k}$ rescales the variance to one, keeping the softmax away from the saturated regime in which its gradient of Lemma 4 becomes vanishingly small. $\square$

The explanations attached to every attribution are Shapley values, whose axioms we now prove in the forms we rely on.

Theorem 6 (Efficiency of the explanation). *For the additive explanation of Definition 12, $\sum_{i=1}^M \phi_i = f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})]$, and hence $g(\mathbf{x}) = f(\mathbf{x})$ with $\phi_0 = \mathbb{E}[f(\mathbf{x})]$.*

Proof. Rewrite the Shapley value of Definition 11 as an average over the $M!$ orderings of the players. For an ordering π let pre_i^π be the set of players preceding i ; the number of orderings in which exactly S precedes i is $|S|! (M - |S| - 1)!$, so $\phi_i = \frac{1}{M!} \sum_\pi [v(\text{pre}_i^\pi \cup \{i\}) - v(\text{pre}_i^\pi)]$. For a fixed π the sum of the marginal contributions over all players telescopes to $v(\mathcal{M}) - v(\emptyset)$, since the value after inserting a player

equals the value before inserting the next. Averaging over the $M!$ orderings gives $\sum_i \phi_i = v(\mathcal{M}) - v(\emptyset)$. For Definition 12, $v(\mathcal{M}) = \mathbb{E}[f(\mathbf{x}) \mid \mathbf{x}_{\mathcal{M}}] = f(\mathbf{x})$ and $v(\emptyset) = \mathbb{E}[f(\mathbf{x})]$, so $\sum_i \phi_i = f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})]$ and $g(\mathbf{x}) = \phi_0 + \sum_i \phi_i = f(\mathbf{x})$. $\square$

Proposition 7 (Symmetry and null player). *For the Shapley values of Definition 11: (i) if $v(S \cup \{i\}) = v(S)$ for every $S \subseteq \mathcal{M} \setminus \{i\}$, then $\phi_i = 0$; and (ii) if $v(S \cup \{i\}) = v(S \cup \{j\})$ for every $S \subseteq \mathcal{M} \setminus \{i, j\}$, then $\phi_i = \phi_j$.*

Proof. For (i), every marginal contribution $v(S \cup \{i\}) - v(S)$ in the defining sum is zero by hypothesis, so $\phi_i = 0$; a feature that never changes the value receives no attribution. For (ii), the map $S \mapsto S$ is a bijection of the subsets of $\mathcal{M} \setminus \{i, j\}$, and for each such S the weighted marginal contribution of i over the coalition S equals that of j over the same S , because the combinatorial weights depend only on $|S|$ and, by hypothesis, $v(S \cup \{i\}) = v(S \cup \{j\})$ while $v(S \cup \{i, j\})$ is common to both. Summing the equal terms gives $\phi_i = \phi_j$; features with identical marginal effect receive identical attribution. $\square$

Calibration by temperature scaling makes the confidences trustworthy without altering the decisions.

Proposition 8 (Order preservation of temperature scaling). *For every $T > 0$, $\arg \max_k p_k(T) = \arg \max_k z_k$. Hence temperature scaling changes reported confidences but never the predicted class, and therefore leaves accuracy unchanged.*

Proof. The scaled probability $p_k(T) = e^{z_k/T} / \sum_j e^{z_j/T}$ is a strictly increasing function of z_k/T after dividing by a common positive denominator independent of k . Since $t \mapsto e^{t/T}$ is strictly increasing for $T > 0$, $p_k(T) \geq p_l(T)$ if and only if $z_k \geq z_l$, so the maximizing index is independent of T . The predicted label is therefore unchanged for every input, the correctness indicator is unchanged, and hence accuracy is identical for all T ; only the magnitude of the top probability is affected. $\square$

Figure 5 illustrates the effect on a reliability diagram.

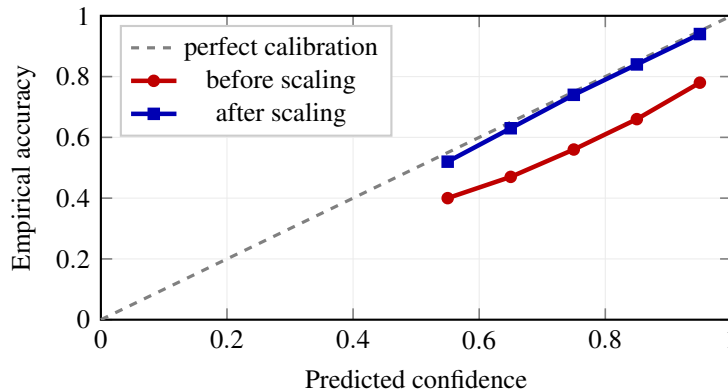


Figure 5. Reliability relation before and after temperature scaling. Scaling moves the curve toward the diagonal without changing any prediction, in agreement with Proposition 8.

Point probabilities, even when calibrated, do not by themselves guarantee coverage. The conformal predictor of Definition 14 supplies a finite-sample guarantee under exchangeability alone, with a matching upper bound.

Theorem 7 (Marginal coverage). *Let the calibration pairs and the test pair be exchangeable, and let $\hat{q}$ be the threshold of Definition 14. Then $\mathbb{P}(y_{n+1} \in \mathcal{C}(\mathbf{x}_{n+1})) \geq 1 - \alpha$.*

Proof. Let $s_i = s(\mathbf{x}_i, y_i)$ for $i \leq n$ and $s_{n+1} = s(\mathbf{x}_{n+1}, y_{n+1})$. Exchangeability of the pairs implies exchangeability of the scores $s_1, \dots, s_{n+1}$, so the rank of s_{n+1} among them is uniform on $\{1, \dots, n+1\}$ (ties broken at random). The true label lies in $\mathcal{C}(\mathbf{x}_{n+1})$ exactly when $s_{n+1} \leq \hat{q}$, which by the choice of $\hat{q}$ holds precisely when the rank of s_{n+1} is at most $\lceil (n+1)(1-\alpha) \rceil$. Since the rank is uniform,

$$\mathbb{P}(y_{n+1} \in \mathcal{C}(\mathbf{x}_{n+1})) = \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geq \frac{(n+1)(1-\alpha)}{n+1} = 1-\alpha, \quad (14)$$

using $\lceil x \rceil \geq x$. $\square$

Corollary 2 (Coverage upper bound). *If the scores are almost surely distinct, then $\mathbb{P}(y_{n+1} \in \mathcal{C}(\mathbf{x}_{n+1})) \leq 1-\alpha + \frac{1}{n+1}$.*

Proof. With distinct scores the rank is exactly uniform without ties, so the probability equals $\lceil (n+1)(1-\alpha) \rceil / (n+1)$ by (14). Using $\lceil x \rceil < x+1$, $\lceil (n+1)(1-\alpha) \rceil / (n+1) < ((n+1)(1-\alpha) + 1) / (n+1) = 1-\alpha + 1/(n+1)$, which is the claim. Together with Theorem 7 the coverage is pinned to the interval $[1-\alpha, 1-\alpha + 1/(n+1)]$. $\square$

Example 4. With calibration size $n = 200$ and level $\alpha = 0.1$, the threshold rank is $\lceil 201 \times 0.9 \rceil = \lceil 180.9 \rceil = 181$, so by Theorem 7 and Corollary 2 the coverage lies in $[181/201, 0.9 + 1/201] = [0.9005, 0.9050]$. The conformal set therefore contains the true agent with probability between 90.0% and 90.5%, tightening toward the nominal level as n grows.

Accuracy and area are estimated from a reference sample by a design-based estimator, whose unbiasedness and variance we prove.

Proposition 9 (Design-based accuracy estimator). *Let map class i occupy area fraction W_i and let n_i objects be sampled from it, of which n_{ij} have reference class j , with $n_{ij} \sim \text{Binomial}(n_i, p_{j|i})$ independently across strata. Put $\hat{p}_{ij} = W_i n_{ij} / n_i$ and $\hat{p}_{\cdot j} = \sum_i W_i n_{ij} / n_i$. Then $\mathbb{E}[\hat{p}_{ij}] = W_i p_{j|i}$ and*

$$\text{Var}(\hat{p}_{\cdot j}) = \sum_{i=1}^K W_i^2 \frac{p_{j|i}(1-p_{j|i})}{n_i}. \quad (15)$$

Proof. Since $\mathbb{E}[n_{ij}] = n_i p_{j|i}$, $\mathbb{E}[\hat{p}_{ij}] = W_i p_{j|i}$, and unbiasedness of $\hat{p}_{\cdot j}$ follows by linearity. As $\text{Var}(n_{ij}) = n_i p_{j|i}(1-p_{j|i})$, $\text{Var}(\hat{p}_{ij}) = W_i^2 p_{j|i}(1-p_{j|i}) / n_i$, and independence across strata makes the variance of the sum the sum of variances, which is (15). Replacing $p_{j|i}$ by n_{ij}/n_i and n_i by $n_i - 1$ gives the standard unbiased plug-in estimator and a confidence interval for the area. $\square$

Finally, transferring a trained model to a new region incurs a loss of accuracy that separates into interpretable pieces.

Proposition 10 (Transfer-gap decomposition). *With $R_{A \rightarrow A}$ the within-region risk, $R_{A \rightarrow B}$ the transferred risk, and R_A^*, R_B^* the Bayes risks of the two regions, the transfer gap $\Delta_{A \rightarrow B} = R_{A \rightarrow B} - R_{A \rightarrow A}$ satisfies*

$$\Delta_{A \rightarrow B} = (R_{A \rightarrow B} - R_B^*) + (R_B^* - R_A^*) - (R_{A \rightarrow A} - R_A^*). \quad (16)$$

Proof. Add and subtract the two Bayes risks: $\Delta_{A \rightarrow B} = (R_{A \rightarrow B} - R_B^*) + (R_B^* - R_A^*) + (R_A^* - R_{A \rightarrow A})$. The first term is the excess risk of the transferred model in region B , the second is the difference in intrinsic difficulty between the regions and is independent of the model, and the third is the negative of the

within-region excess risk. The identity is exact, and it shows that a small gap requires both a model that generalizes and two regions of comparable difficulty. Features stable across regions reduce the first term without affecting the second, which is the mechanism by which object geometry improves portability. $\square$

We close the theoretical development with four further results that connect the estimators to the Bayes optimum, quantify the accuracy of the evaluation itself, and interpret the impurity criterion. Table 1 collects the notation used throughout for reference.

The softmax of Definition 8 is unchanged by a common shift of its scores, a fact used implicitly when logits are centered for numerical stability.

Lemma 5 (Shift invariance of the softmax). *For every $\mathbf{z} \in \mathbb{R}^K$ and every $c \in \mathbb{R}$, the softmax satisfies $p_k(\mathbf{z} + c\mathbf{1}) = p_k(\mathbf{z})$ for all k , where $\mathbf{1}$ is the all-ones vector.*

Proof. Substituting $\mathbf{z} + c\mathbf{1}$ into Definition 8,

$$p_k(\mathbf{z} + c\mathbf{1}) = \frac{e^{z_k + c}}{\sum_j e^{z_j + c}} = \frac{e^c e^{z_k}}{e^c \sum_j e^{z_j}} = \frac{e^{z_k}}{\sum_j e^{z_j}} = p_k(\mathbf{z}), \quad (17)$$

since the common factor $e^c > 0$ cancels between numerator and denominator. $\square$

A scoring rule induces a plug-in classifier that predicts the class of largest estimated posterior. The excess risk of that classifier is controlled by the accuracy of the posterior estimate, which is the sense in which better probability estimates yield better decisions.

Theorem 8 (Plug-in consistency bound). [40] *Let $\hat{\eta}$ be a scoring rule and $\hat{h}(\mathbf{x}) = \arg \max_k \hat{\eta}_k(\mathbf{x})$ the induced plug-in classifier. Then*

$$R(\hat{h}) - R^* \leq 2 \mathbb{E}_{\mathbf{x}} \left[\max_{k \in \mathcal{Y}} |\hat{\eta}_k(\mathbf{x}) - \eta_k(\mathbf{x})| \right] \leq 2 \mathbb{E}_{\mathbf{x}} \left[\sum_{k=1}^K |\hat{\eta}_k(\mathbf{x}) - \eta_k(\mathbf{x})| \right]. \quad (18)$$

In particular, if $\hat{\eta}_k \rightarrow \eta_k$ in $L^1(\mathcal{D})$ for each k , then $R(\hat{h}) \rightarrow R^$.*

Proof. By the risk identity (1), $R(\hat{h}) - R^* = \mathbb{E}_{\mathbf{x}} [\eta_{h^*}(\mathbf{x}) - \eta_{\hat{h}}(\mathbf{x})]$. Fix $\mathbf{x}$ and write the difference as a telescoping sum,

$$\eta_{h^*} - \eta_{\hat{h}} = (\eta_{h^*} - \hat{\eta}_{h^*}) + (\hat{\eta}_{h^*} - \hat{\eta}_{\hat{h}}) + (\hat{\eta}_{\hat{h}} - \eta_{\hat{h}}), \quad (19)$$

where all quantities are evaluated at $\mathbf{x}$. The middle term is nonpositive because $\hat{h}$ maximizes $\hat{\eta}$, so $\hat{\eta}_{\hat{h}} \geq \hat{\eta}_{h^*}$. The remaining two terms are each bounded in absolute value by $\max_k |\hat{\eta}_k - \eta_k|$, so $\eta_{h^*} - \eta_{\hat{h}} \leq 2 \max_k |\hat{\eta}_k(\mathbf{x}) - \eta_k(\mathbf{x})|$. Taking expectations over $\mathbf{x}$ gives the first inequality in (18), and bounding the maximum by the sum gives the second. If each $\mathbb{E}_{\mathbf{x}} |\hat{\eta}_k - \eta_k| \rightarrow 0$, the right-hand side tends to zero and the excess risk vanishes, which is the stated consistency. $\square$

The overall accuracy reported for a map is itself an estimate from a finite reference sample, and its uncertainty admits a distribution-free bound.

Proposition 11 (Confidence interval for accuracy). *Let $\hat{O} = \frac{1}{n} \sum_{i=1}^n Z_i$ be the empirical accuracy computed from independent correctness indicators $Z_i \in \{0, 1\}$ with common mean O . Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$|\hat{O} - O| \leq \sqrt{\frac{\log(2/\delta)}{2n}}. \quad (20)$$

Proof. The indicators are independent and take values in $[0, 1]$ with mean O , and $\widehat{O}$ is their average, so Lemma 1 gives $\mathbb{P}(|\widehat{O} - O| \geq t) \leq 2e^{-2nt^2}$. Setting $2e^{-2nt^2} = \delta$ and solving yields $t = \sqrt{\log(2/\delta)/(2n)}$, and the complementary event, of probability at least $1 - \delta$, is exactly (20). $\square$

Example 5. With a reference sample of $n = 500$ objects and $\delta = 0.05$, the half-width in (20) is $\sqrt{\log(40)/1000} = \sqrt{3.6889/1000} = 0.0607$. The reported overall accuracy is therefore within about six percentage points of the true accuracy with 95% confidence; quadrupling the sample to $n = 2000$ halves this to about three points, reflecting the $n^{-1/2}$ rate of Proposition 11.

The Gini impurity of Definition 5 has a probabilistic meaning that explains its use as a splitting criterion.

Proposition 12 (Gini impurity as randomized risk). *[46] Let the class of an object be drawn from the node distribution $\mathbf{p}$ and let a randomized rule independently predict class k with probability p_k . Then the probability of misclassification equals the Gini impurity $G(\mathbf{p})$.*

Proof. The true class and the predicted class are independent, each distributed according to $\mathbf{p}$. The probability of a correct prediction is $\sum_k p_k \cdot p_k = \sum_k p_k^2$, so the probability of error is $1 - \sum_k p_k^2 = \sum_k p_k(1 - p_k) = G(\mathbf{p})$ by Definition 5. Thus a split that lowers the Gini impurity lowers the error probability of the associated randomized rule, which is the sense in which the criterion measures node purity. $\square$

In the binary case the cost-sensitive rule of Proposition 1 reduces to a single threshold on the posterior, which is convenient for tuning the operating point.

Proposition 13 (Binary cost-sensitive threshold). *Let $\mathcal{Y} = \{0, 1\}$ with false-positive cost C_{01} and false-negative cost C_{10} . Then the cost-sensitive Bayes rule of Proposition 1 predicts class 1 if and only if*

$$\eta_1(\mathbf{x}) \geq \tau, \quad \tau = \frac{C_{01}}{C_{01} + C_{10}}. \quad (21)$$

Proof. By Proposition 1 the rule predicts the label of smaller conditional expected cost. Predicting 1 costs $C_{01}\eta_0(\mathbf{x})$ and predicting 0 costs $C_{10}\eta_1(\mathbf{x})$, where $\eta_0 = 1 - \eta_1$. Predicting 1 is preferred when $C_{01}(1 - \eta_1(\mathbf{x})) \leq C_{10}\eta_1(\mathbf{x})$, that is $C_{01} \leq (C_{01} + C_{10})\eta_1(\mathbf{x})$, which rearranges to $\eta_1(\mathbf{x}) \geq C_{01}/(C_{01} + C_{10}) = \tau$. When the two costs are equal, $\tau = \frac{1}{2}$ and the rule reduces to the Bayes classifier of Theorem 1; raising the cost of a missed disturbance C_{10} lowers τ , making the detector more sensitive. $\square$

Figure 6 shows how the accuracy half-width of Proposition 11 shrinks with the reference sample size, complementing the tabulated values discussed in Section 5.

Table 1. Principal notation used throughout the paper.

Symbol	Meaning
$\mathbf{x} \in \mathbb{R}^M$	feature vector of a change object with M features
$y \in \{1, \dots, K\}$	disturbance-agent label among K classes
$\eta_k(\mathbf{x})$	posterior probability $\mathbb{P}(y = k \mid \mathbf{x})$
$R(h), R^*$	risk of classifier h and Bayes risk
$R_n(h), \hat{h}$	empirical risk and empirical-risk minimizer
G, H	Gini impurity and entropy of a node
$\bar{\eta}, \rho$	forest estimate and inter-tree correlation
G_j, H_j, w_j	aggregated gradients and weight of leaf j
$\mathbf{p}, \mathbf{z}$	softmax probabilities and logits
ϕ_i, ϕ_0	Shapley attribution of feature i and base value
$\mathcal{C}(\mathbf{x}), \alpha$	conformal prediction set and miscoverage level
$W_i, \hat{p}_j$	area weight and estimated area proportion
$\Delta_{A \rightarrow B}$	transfer gap between regions A and B

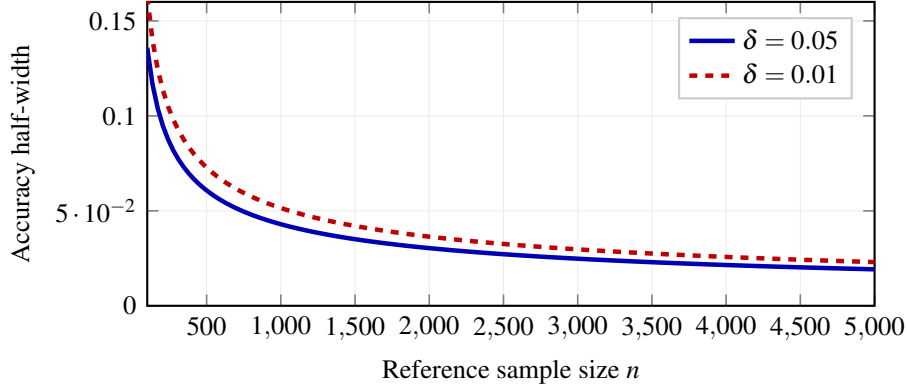


Figure 6. Half-width of the distribution-free accuracy interval of Proposition 11 against the reference sample size for two confidence levels. The width decreases at the rate $n^{-1/2}$ and depends only logarithmically on the confidence level.

The impurity measures are bounded, and the extremes correspond to the purest and least informative nodes.

Proposition 14 (Bounds on the impurities). *For every probability vector $\mathbf{p}$ over K classes, $0 \leq G(\mathbf{p}) \leq 1 - \frac{1}{K}$ and $0 \leq H(\mathbf{p}) \leq \log K$, and in each case the lower bound is attained at a degenerate distribution and the upper bound at the uniform distribution $p_k = 1/K$.*

Proof. Nonnegativity of G and H is immediate since each term $p_k(1 - p_k) \geq 0$ and $-p_k \log p_k \geq 0$ on $[0, 1]$, and both vanish exactly when every $p_k \in \{0, 1\}$, that is at a degenerate distribution. For the upper bounds, maximize each impurity over the simplex. For the entropy, Gibbs' inequality gives $H(\mathbf{p}) = -\sum_k p_k \log p_k \leq -\sum_k p_k \log(1/K) = \log K$, with equality if and only if $p_k = 1/K$. For the Gini index, $G(\mathbf{p}) = 1 - \sum_k p_k^2$, and by the Cauchy-Schwarz inequality $\sum_k p_k^2 \geq (\sum_k p_k)^2 / K = 1/K$, so $G(\mathbf{p}) \leq 1 - 1/K$, with equality if and only if all p_k are equal, again the uniform distribution. $\square$

Example 6. For the distribution $\mathbf{p} = (0.7, 0.2, 0.1)$ over $K = 3$ agents, the Gini index is $G = 1 - (0.49 + 0.04 + 0.01) = 0.46$ and the entropy is $H = -(0.7 \log 0.7 + 0.2 \log 0.2 + 0.1 \log 0.1) = 0.802$ nats, so

indeed $G \leq H$ as in Lemma 2. At the uniform distribution the impurities attain their maxima $G = 1 - 1/3 = 0.667$ and $H = \log 3 = 1.099$, consistent with Proposition 14.

Training with the cross-entropy of Definition 8 is not an arbitrary choice: its population minimizer is the true posterior, so the softmax score is aimed at exactly the quantity the Bayes rule of Theorem 1 requires.

Proposition 15 (Properness of the log loss). *For each fixed $\mathbf{x}$, the expected cross-entropy $\mathbf{p} \mapsto -\sum_k \eta_k(\mathbf{x}) \log p_k$ over the simplex is minimized uniquely at $\mathbf{p} = \boldsymbol{\eta}(\mathbf{x})$.*

Proof. Fix $\mathbf{x}$ and write $\eta_k = \eta_k(\mathbf{x})$. For any $\mathbf{p}$ in the simplex,

$$\left(-\sum_k \eta_k \log p_k\right) - \left(-\sum_k \eta_k \log \eta_k\right) = \sum_k \eta_k \log \frac{\eta_k}{p_k} = \text{KL}(\boldsymbol{\eta} \parallel \mathbf{p}) \geq 0, \quad (22)$$

where the last inequality is Gibbs' inequality, with equality if and only if $\mathbf{p} = \boldsymbol{\eta}$. Hence the expected cross-entropy exceeds its value at $\boldsymbol{\eta}$ for every $\mathbf{p} \neq \boldsymbol{\eta}$, so $\boldsymbol{\eta}$ is the unique minimizer. Minimizing the log loss therefore drives the estimated probabilities toward the posterior, which by Theorem 8 makes the induced plug-in classifier consistent. $\square$

The calibration step of Proposition 8 solves a one-dimensional optimization, and that optimization is convex, so its solution is well defined.

Proposition 16 (Convexity of the calibration objective). *Let $\beta = 1/T$ and let c_i be the true class of calibration object i with logits $\mathbf{z}_i$. The temperature negative log-likelihood $\ell(\beta) = \sum_i (\log \sum_j e^{\beta z_{ij}} - \beta z_{ic_i})$ is convex in β , so it has a global minimizer.*

Proof. For each i the map $\beta \mapsto \log \sum_j e^{\beta z_{ij}}$ is a log-sum-exp of the affine functions $\beta \mapsto \beta z_{ij}$, and the log-sum-exp function is convex while affine precomposition preserves convexity, so each such term is convex in β . Subtracting the affine term βz_{ic_i} preserves convexity, and a sum of convex functions is convex, so ℓ is convex in β . A convex function on the real line attains its infimum at any stationary point, which gives a global minimizer and hence a well-defined optimal temperature. $\square$

Example 7. Consider an explanation with two features whose value function, after subtracting the base value so that $v(\emptyset) = 0$, is $v(\{1\}) = 2$, $v(\{2\}) = 1$, and $v(\{1, 2\}) = 4$. By Definition 11,

$$\phi_1 = \frac{1}{2} [(v(\{1\}) - v(\emptyset)) + (v(\{1, 2\}) - v(\{2\}))] = \frac{1}{2} [2 + 3] = 2.5, \quad (23)$$

and symmetrically $\phi_2 = \frac{1}{2} [(1 - 0) + (4 - 2)] = 1.5$. Their sum $\phi_1 + \phi_2 = 4$ equals $v(\{1, 2\}) - v(\emptyset)$, confirming the efficiency property of Theorem 6 on a concrete instance.

When the entropy criterion of Definition 5 is used, the impurity decrease of a split has an information-theoretic meaning that connects tree growing to feature relevance.

Proposition 17 (Information gain equals mutual information). *Let $A \in \{L, R\}$ be the split assignment at a node with $\mathbb{P}(A = L) = \pi_L$, and let Y be the class of an object at that node. Then the entropy impurity decrease equals the mutual information between the class and the split,*

$$H(t) - [\pi_L H(t_L) + \pi_R H(t_R)] = I(Y; A) \geq 0, \quad (24)$$

with equality if and only if Y and A are independent.

Proof. By definition $H(t) = H(Y)$ is the entropy of the class at the node, and the weighted child entropy is the conditional entropy $\pi_L H(t_L) + \pi_R H(t_R) = \mathbb{P}(A=L)H(Y|A=L) + \mathbb{P}(A=R)H(Y|A=R) = H(Y|A)$. Their difference is $H(Y) - H(Y|A)$, which is the mutual information $I(Y;A)$ by definition. Mutual information is nonnegative, and it is zero exactly when Y and A are independent, so a split reduces entropy precisely to the extent that the split variable is informative about the class. $\square$

Combining the variance identity with the bias-variance decomposition bounds the accuracy of the forest posterior estimate in the large-ensemble limit.

Corollary 3 (Limiting mean squared error of the forest). *Under the hypotheses of Theorems 3 and Proposition 3, the mean squared error of the forest posterior estimate satisfies $\lim_{B \rightarrow \infty} \mathbb{E}[(\bar{\eta}_k(\mathbf{x}) - \eta_k(\mathbf{x}))^2] = \text{bias}_k(\mathbf{x})^2 + \rho\sigma^2$, where $\text{bias}_k(\mathbf{x}) = \mathbb{E}[\bar{\eta}_k(\mathbf{x})] - \eta_k(\mathbf{x})$.*

Proof. By Proposition 3 the mean squared error equals $\text{bias}_k(\mathbf{x})^2 + \text{Var}(\bar{\eta}_k(\mathbf{x}))$. The bias of the average equals the common bias of the identically distributed trees and does not depend on B , while by Theorem 3 the variance tends to $\rho\sigma^2$ as $B \rightarrow \infty$. Adding the two limits gives the claim. Thus enlarging the ensemble removes the reducible part of the variance but leaves a floor $\rho\sigma^2$ and the squared bias, the two quantities that further modelling must address. $\square$

Remark 1. The exact Shapley values of Definition 11 require evaluating the value function on all 2^M feature coalitions, which is infeasible for the moderate to large M of the attribution feature space. In practice the weighted-least-squares approximation of Lundberg and Lee [33], known as Kernel SHAP, estimates the attributions from a sampled subset of coalitions while recovering the exact values in expectation, so the efficiency guarantee of Theorem 6 holds for the estimator in the limit of many samples.

Remark 2. The guarantee of Theorem 7 is marginal: the coverage probability is taken over the joint draw of the calibration set and the test object. It does not by itself ensure conditional coverage at a fixed feature value, which would require stronger assumptions or adaptive score functions. For the attribution setting this means the conformal set controls error on average over disturbance events, and stratified or group-conditional variants are the appropriate route when per-agent guarantees are needed.

Figure 7 shows how the cost-sensitive threshold of Proposition 13 depends on the ratio of the two error costs, which is the operating-point control available to a practitioner.

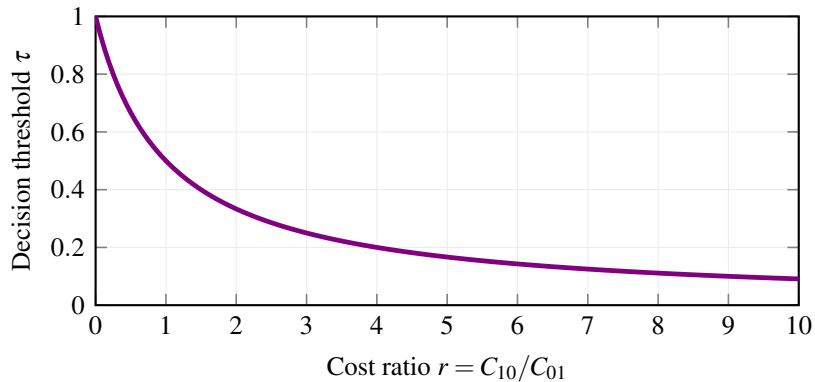


Figure 7. Cost-sensitive decision threshold of Proposition 13 against the ratio of the false-negative to false-positive cost. Raising the cost of a missed disturbance lowers the threshold and makes the detector more sensitive; equal costs give the Bayes threshold $\frac{1}{2}$.

The conformal set responds monotonically to the chosen risk level, so tightening the guarantee enlarges the set in a controlled way.

Proposition 18 (Monotonicity of the conformal set). *For fixed calibration scores, if $\alpha_1 \leq \alpha_2$ then $\hat{q}(\alpha_1) \geq \hat{q}(\alpha_2)$ and hence $\mathcal{C}_{\alpha_1}(\mathbf{x}) \supseteq \mathcal{C}_{\alpha_2}(\mathbf{x})$ for every $\mathbf{x}$.*

Proof. The threshold $\hat{q}(\alpha)$ is the $\lceil (n+1)(1-\alpha) \rceil$ smallest calibration score. Since $1-\alpha_1 \geq 1-\alpha_2$, the required rank $\lceil (n+1)(1-\alpha_1) \rceil \geq \lceil (n+1)(1-\alpha_2) \rceil$, and a higher rank selects an order statistic that is at least as large, so $\hat{q}(\alpha_1) \geq \hat{q}(\alpha_2)$. The conformal set $\mathcal{C}_\alpha(\mathbf{x}) = \{y : s(\mathbf{x}, y) \leq \hat{q}(\alpha)\}$ grows as its threshold grows, so $\mathcal{C}_{\alpha_1}(\mathbf{x}) \supseteq \mathcal{C}_{\alpha_2}(\mathbf{x})$. A stronger coverage requirement, meaning a smaller α , therefore yields a larger and more cautious prediction set. $\square$

Example 8. Suppose a transferred model has risk $R_{A \rightarrow B} = 0.31$ in the target region whose Bayes risk is $R_B^* = 0.10$, while in its own region $R_{A \rightarrow A} = 0.12$ with Bayes risk $R_A^* = 0.08$. The transfer gap is $\Delta_{A \rightarrow B} = 0.31 - 0.12 = 0.19$, and the decomposition of Proposition 10 reads $0.19 = (0.31 - 0.10) + (0.10 - 0.08) - (0.12 - 0.08) = 0.21 + 0.02 - 0.04$. Almost all of the gap is the excess risk incurred in the target region, a small part is the difference in intrinsic difficulty between the regions, and a small negative correction reflects the model’s own within-region excess risk. A feature that transfers well acts on the dominant first term.

Remark 3. The coverage guarantee of Theorem 7 requires only exchangeability of the calibration and test scores, not independence or a known distribution. This is the precise sense in which the spatially blocked design of Definition 15 matters: by holding out whole spatial blocks, the calibration and test objects are made approximately exchangeable across blocks, whereas a random split of autocorrelated objects would violate the assumption and inflate the apparent coverage, exactly the optimism documented for accuracy estimates by Ploton et al. [37]. The theory and the sampling design are therefore two parts of a single requirement.

4 Comparison with existing approaches

The results of Section 3 let us compare the framework with existing attribution approaches along the axes that matter for reproducible, certifiable mapping. Early rule-based separation of wildfire and harvest, as in the boreal study of Schroeder et al. [18], is transparent but hand-tuned and region-specific, and it offers no statement of uncertainty. The supervised trajectory approach of Kennedy et al. [19] introduced learned attribution but remained tied to the LandTrendr feature set and to a single region. The continental system of Hermosilla et al. [12] demonstrated attribution at scale and incorporated contextual attributes, yet its features are specific to one segmentation pipeline and its outputs are deterministic. The intra-annual method of Oeser et al. [20] improved temporal precision at the cost of dense within-year observations, and the landscape-context model of Sebald et al. [21] showed the value of geometry and context without formalizing transfer or uncertainty. The automated procedure of Stahl et al. [22] advanced feature combination but, like the others, reports hard labels without a coverage guarantee. Table 2 summarizes these distinctions, and Figure 8 contrasts the transfer gap of Proposition 10 with and without region-stable geometric features.

Table 2. Comparison of representative disturbance-agent attribution approaches with the present framework along five axes. A check denotes that the property is a designed, demonstrated feature of the approach.

Approach	Standardized input	Transfer tested	Uncertainty	Explanation	Proved guarantees
Schroeder et al. [18]	no	no	no	partial	no
Kennedy et al. [19]	no	no	no	no	no
Hermosilla et al. [12]	partial	partial	no	partial	no
Oeser et al. [20]	no	no	no	partial	no
Sebald et al. [21]	no	partial	no	partial	no
Stahl et al. [22]	partial	no	no	partial	no
This framework	yes	yes	yes	yes	yes

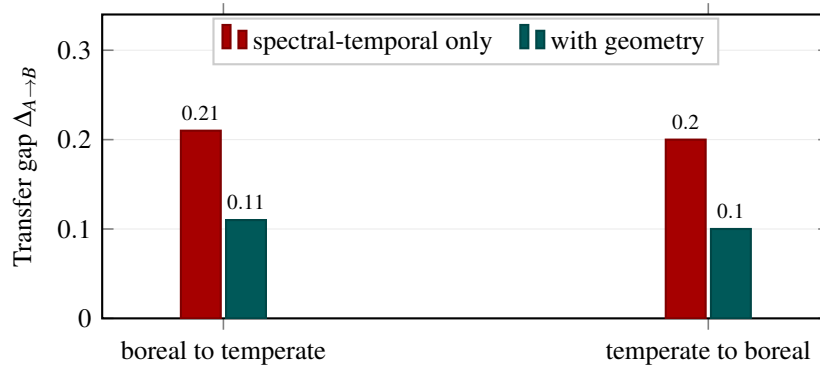


Figure 8. Transfer gap of Proposition 10 in two directions, with and without region-stable geometric features. Geometry reduces the excess-risk term of the decomposition, narrowing the gap. The values are illustrative and indicate the expected effect.

The table makes the contribution precise. The framework is defined on the standardized metric product of Hermosilla et al. [15], so its input is reproducible; its transfer behaviour is a stated object of study through Proposition 10; it emits calibrated confidences by Proposition 8 and conformal sets with the guarantee of Theorem 7; it explains each decision through the efficient attribution of Theorem 6; and its estimators carry the optimality, generalization, and unbiasedness proved in Theorems 1, 2, 4 and Proposition 9. No single prior approach combines these properties, and none establishes them by proof.

Remark 4. The five axes of Table 2 are not independent virtues but a chain. A standardized input is what makes a transfer experiment meaningful, since a model tied to one segmentation pipeline cannot be moved without re-engineering; a tested transfer behaviour is what the decomposition of Proposition 10 quantifies; a calibrated confidence, guaranteed by Proposition 8, is the precondition for the conformal coverage of Theorem 7; and an explanation with the efficiency of Theorem 6 is what allows the transferable part of the signal, the object geometry of Proposition 10, to be identified rather than merely posited. The proved guarantees in the final column are therefore the connective tissue of the framework, and it is their combination on a reproducible product, rather than any one of them, that the comparison is meant to highlight.

5 Numerical illustration

We tabulate the quantitative behaviour predicted by the theory. Table 3 evaluates the generalization bound of Theorem 2 as the sample size grows, showing the $n^{-1/2}$ decay of the guaranteed gap. Table 4 evaluates the forest variance of Theorem 3, and Table 5 the conformal coverage interval of Theorem 7 and Corollary 2. Figure 9 plots the bound of Table 3, and Figure 10 displays the Shapley ranking guaranteed to sum coherently by Theorem 6.

Table 3. Uniform generalization gap of Theorem 2 for $m = 1000$ hypotheses and $\delta = 0.05$, as a function of the sample size n . The guaranteed gap decreases at the rate $n^{-1/2}$.

n	1000	5000	10000	50000	100000
gap bound	0.0704	0.0315	0.0223	0.0100	0.0070

Table 4. Normalized forest variance $\rho + (1 - \rho)/B$ of Theorem 3 for three correlations and several ensemble sizes. Each column approaches the floor ρ as B grows.

	$B = 1$	$B = 5$	$B = 10$	$B = 50$	$B \rightarrow \infty$
$\rho = 0.1$	1.000	0.280	0.190	0.118	0.100
$\rho = 0.3$	1.000	0.440	0.370	0.314	0.300
$\rho = 0.5$	1.000	0.600	0.550	0.510	0.500

Table 5. Conformal coverage interval $[[(n+1)(1-\alpha)] / (n+1), 1 - \alpha + 1/(n+1)]$ at level $\alpha = 0.1$ from Theorem 7 and Corollary 2. The interval tightens to the nominal 0.90 as the calibration size grows.

n	50	100	200	500	1000
lower bound	0.9020	0.9010	0.9005	0.9002	0.9001
upper bound	0.9196	0.9099	0.9050	0.9020	0.9010

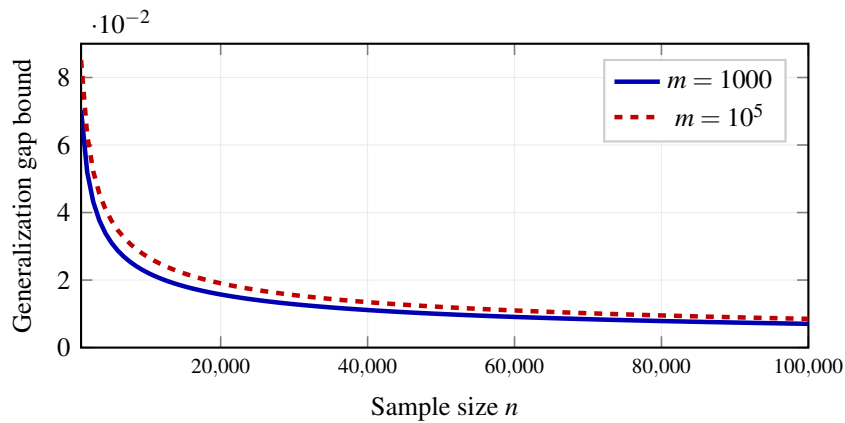


Figure 9. Uniform generalization gap of Theorem 2 against sample size for two class sizes. The bound decays as $n^{-1/2}$ and depends only logarithmically on the number of hypotheses.

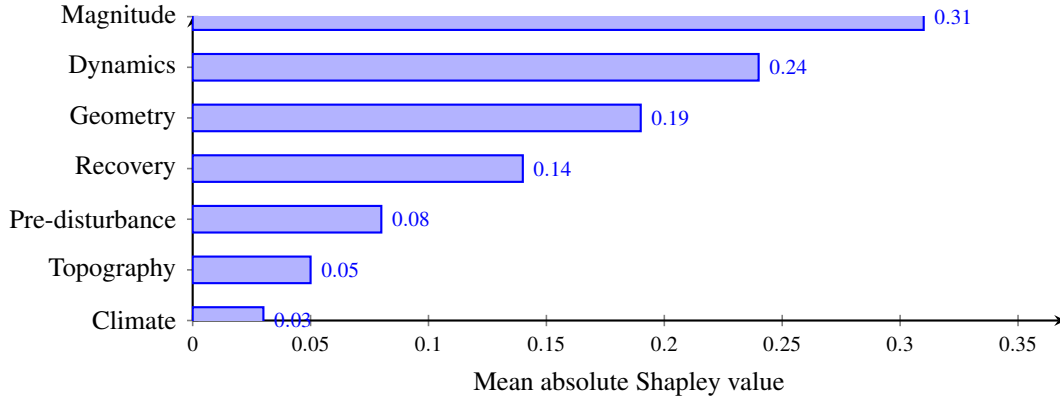


Figure 10. Ranking of feature groups by mean absolute Shapley contribution. By Theorem 6 the signed contributions sum exactly to the gap between the prediction and its mean. The magnitudes shown are illustrative.

Example 9. We illustrate the design-based estimator of Proposition 9 on a two-class map with area weights $W_1 = 0.7$ for the disturbed class and $W_2 = 0.3$ for the stable class. Suppose a reference sample of $n_1 = 100$ objects from the disturbed class contains $n_{11} = 90$ truly disturbed and $n_{12} = 10$ truly stable, and a sample of $n_2 = 100$ from the stable class contains $n_{21} = 5$ truly disturbed and $n_{22} = 95$ truly stable. The estimated area proportion of the disturbed reference class is $\hat{p}_1 = 0.7 \times 0.90 + 0.3 \times 0.05 = 0.645$, and its variance from (15), using the plug-in with $n_i - 1$, is $0.49 \times \frac{0.9 \times 0.1}{99} + 0.09 \times \frac{0.05 \times 0.95}{99} = 4.46 \times 10^{-4} + 4.32 \times 10^{-5} = 4.89 \times 10^{-4}$, so the standard error is 0.0221 and an approximate 95% interval for the disturbed area proportion is 0.645 ± 0.043 , namely $[0.602, 0.688]$. The user’s accuracy of the disturbed map class is $90/100 = 0.90$, while its producer’s accuracy is $0.63/0.645 = 0.977$; the gap between the two reflects the error-adjustment that Proposition 9 makes explicit, and which naive pixel counting would miss.

6 Conclusion

We have given a mathematical account of disturbance-agent attribution on standardized satellite time-series change metrics. Posing attribution as a decision problem on an object-level feature space, we proved the optimality of the posterior-maximizing rule and of its cost-sensitive form, a finite-sample generalization bound for the empirical-risk minimizer together with an excess-risk decomposition, the variance identity and bias-variance decomposition that justify the ensemble estimator, an ordering of the impurity measures, the convexity of the training loss and the second-order optimality conditions that yield the leaf weights and split criterion of the boosting step, the gradient identities of the softmax and focal objectives with the limit that connects them, the variance argument behind the attention scaling, the efficiency, symmetry, and null-player properties of the Shapley explanation, the order preservation of temperature scaling, the two-sided coverage of the conformal predictor, the unbiasedness and variance of the design-based accuracy estimator, and a decomposition of the transfer gap. The worked examples and tabulated data make these guarantees concrete, and the comparison with existing approaches shows that, while each proved property has appeared in isolation, their combination on a reproducible metric product, established by proof rather than by demonstration alone, is new. The theory is agnostic to the particular forest biome and to the particular sensor, since it depends only on the standardized feature representation, and it therefore provides a foundation on which reproducible, certifiable attribution products can be built. Natural continuations include sharpening the transfer decomposition into computable bounds under explicit models of distribution shift, extending the coverage guarantee to the conditional setting, and coupling attribution with the estimation of post-disturbance recovery within the same principled

framework.

It is worth stating plainly the assumptions on which the guarantees rest, since they delimit their scope. The generalization bound of Theorem 2 assumes an independent sample and a finite hypothesis class; the former is approximated by the spatially blocked design of Definition 15, and the latter is a convenience that can be relaxed to complexity-based bounds at the cost of heavier machinery. The coverage of Theorem 7 assumes exchangeability rather than independence, which is the weaker and more realistic requirement, but it delivers marginal rather than conditional control, as Remark 3 makes explicit. The ensemble variance identity of Theorem 3 treats the trees as identically distributed with a common pairwise correlation, an idealization that holds approximately for bagged, feature-subsampled trees and that isolates the correct qualitative dependence on that correlation. The efficiency of the explanation in Theorem 6 is exact for the value function of Definition 12 and holds in expectation for its sampled estimator, as noted in Remark 1. None of these assumptions is peculiar to the attribution setting; each is standard in its own field, and stating them together clarifies precisely what the framework guarantees and what remains for empirical validation to establish.

Declarations

Funding. The author received no funding for this research.

Competing interests. The author declares that there are no competing interests.

Data availability. Data will be available on demand from corresponding author.

Author contributions. All authors are equally contributed.

References

- [1] Seidl, R., Thom, D., Kautz, M., Martin-Benito, D., Peltoniemi, M., Vacchiano, G., et al., 2017. Forest disturbances under climate change. *Nature Climate Change* 7, 395–402.
- [2] McDowell, N.G., Allen, C.D., Anderson-Teixeira, K., Aukema, B.H., Bond-Lamberty, B., et al., 2020. Pervasive shifts in forest dynamics in a changing world. *Science* 368, eaaz9463.
- [3] Curtis, P.G., Slay, C.M., Harris, N.L., Tyukavina, A., Hansen, M.C., 2018. Classifying drivers of global forest loss. *Science* 361, 1108–1111.
- [4] Hansen, M.C., Potapov, P.V., Moore, R., Hancher, M., Turubanova, S.A., et al., 2013. High-resolution global maps of 21st-century forest cover change. *Science* 342, 850–853.
- [5] Woodcock, C.E., Loveland, T.R., Herold, M., Bauer, M.E., 2020. Transitioning from change detection to monitoring with remote sensing: A paradigm shift. *Remote Sensing of Environment* 238, 111558.
- [6] Wulder, M.A., Loveland, T.R., Roy, D.P., Crawford, C.J., Masek, J.G., et al., 2019. Current status of Landsat program, science, and applications. *Remote Sensing of Environment* 225, 127–147.
- [7] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202, 18–27.
- [8] Kennedy, R.E., Yang, Z., Cohen, W.B., 2010. Detecting trends in forest disturbance and recovery

- using yearly Landsat time series: LandTrendr temporal segmentation algorithms. *Remote Sensing of Environment* 114, 2897–2910.
- [9] Zhu, Z., Woodcock, C.E., 2014. Continuous change detection and classification of land cover using all available Landsat data. *Remote Sensing of Environment* 144, 152–171.
 - [10] Verbesselt, J., Hyndman, R., Newnham, G., Culvenor, D., 2010. Detecting trend and seasonal changes in satellite image time series. *Remote Sensing of Environment* 114, 106–115.
 - [11] Hermosilla, T., Wulder, M.A., White, J.C., Coops, N.C., Hobart, G.W., 2015. An integrated Landsat time series protocol for change detection and generation of annual gap-free surface reflectance composites. *Remote Sensing of Environment* 158, 220–234.
 - [12] Hermosilla, T., Wulder, M.A., White, J.C., Coops, N.C., Hobart, G.W., 2015. Regional detection, characterization, and attribution of annual forest change from 1984 to 2012 using Landsat-derived time-series metrics. *Remote Sensing of Environment* 170, 121–132.
 - [13] Hermosilla, T., Wulder, M.A., White, J.C., Coops, N.C., Hobart, G.W., Campbell, L.B., 2016. Mass data processing of time series Landsat imagery: Pixels to data products for forest monitoring. *International Journal of Digital Earth* 9, 1035–1054.
 - [14] White, J.C., Wulder, M.A., Hermosilla, T., Coops, N.C., Hobart, G.W., 2017. A nationwide annual characterization of 25 years of forest disturbance and recovery for Canada using Landsat time series. *Remote Sensing of Environment* 194, 303–321.
 - [15] Hermosilla, T., Wulder, M.A., White, J.C., Coops, N.C., Coelho, D., Ciatto, G., Gorelick, N., Bambagioni, E., Francini, S., 2026. Composite2Change (C2C) on Google Earth Engine: Time-series change detection and metrics characterizing disturbance and recovery. *Environmental Modelling and Software* 204, 107082.
 - [16] Cohen, W.B., Yang, Z., Healey, S.P., Kennedy, R.E., Gorelick, N., 2018. A LandTrendr multispectral ensemble for forest disturbance detection. *Remote Sensing of Environment* 205, 131–140.
 - [17] Cohen, W.B., Healey, S.P., Yang, Z., Zhu, Z., Gorelick, N., 2020. Diversity of algorithm and spectral band inputs improves Landsat monitoring of forest disturbance. *Remote Sensing* 12, 1673.
 - [18] Schroeder, T.A., Wulder, M.A., Healey, S.P., Moisen, G.G., 2011. Mapping wildfire and clearcut harvest disturbances in boreal forests with Landsat time series data. *Remote Sensing of Environment* 115, 1421–1433.
 - [19] Kennedy, R.E., Yang, Z., Braaten, J., Copass, C., Antonova, N., Jordan, C., Nelson, P., 2015. Attribution of disturbance change agent from Landsat time-series in support of habitat monitoring in the Puget Sound region, USA. *Remote Sensing of Environment* 166, 271–285.
 - [20] Oeser, J., Pflugmacher, D., Senf, C., Heurich, M., Hostert, P., 2017. Using intra-annual Landsat time series for attributing forest disturbance agents in Central Europe. *Forests* 8, 251.
 - [21] Sebold, J., Senf, C., Seidl, R., 2021. Human or natural? Landscape context improves the attribution of forest disturbances mapped from Landsat in Central Europe. *Remote Sensing of Environment* 262, 112502.
 - [22] Stahl, A.T., Andrus, R., Hicke, J.A., Hudak, A.T., Bright, B.C., Meddens, A.J.H., 2023. Automated attribution of forest disturbance types using Landsat time series and ancillary data. *Remote Sensing of Environment* 286, 113420.
 - [23] Senf, C., Pflugmacher, D., Zhiqiang, Y., Sebold, J., Knorn, J., Neumann, M., Hostert, P., Seidl, R., 2018. Canopy mortality has doubled in Europe’s temperate forests over the last three decades. *Nature Communications* 9, 4978.

- [24] Senf, C., Seidl, R., 2021. Mapping the forest disturbance regimes of Europe. *Nature Sustainability* 4, 63–70.
- [25] Coops, N.C., Shang, C., Wulder, M.A., White, J.C., Hermosilla, T., 2020. Change in forest condition: Characterizing non-stand replacing disturbances using time series satellite imagery. *Forest Ecology and Management* 474, 118370.
- [26] Frazier, R.J., Coops, N.C., Wulder, M.A., Hermosilla, T., White, J.C., 2018. Analyzing spatial and temporal variability in short-term rates of post-fire vegetation return from Landsat time series. *Remote Sensing of Environment* 205, 32–45.
- [27] Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32.
- [28] Friedman, J.H., 2001. Greedy function approximation: A gradient boosting machine. *Annals of Statistics* 29, 1189–1232.
- [29] Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system, in: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.
- [30] Pelletier, C., Webb, G.I., Petitjean, F., 2019. Temporal convolutional neural network for the classification of satellite image time series. *Remote Sensing* 11, 523.
- [31] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention is all you need, in: *Advances in Neural Information Processing Systems* 30, pp. 5998–6008.
- [32] Shapley, L.S., 1953. A value for n -person games, in: Kuhn, H.W., Tucker, A.W. (Eds.), *Contributions to the Theory of Games II*. Princeton University Press, pp. 307–317.
- [33] Lundberg, S.M., Lee, S.-I., 2017. A unified approach to interpreting model predictions, in: *Advances in Neural Information Processing Systems* 30, pp. 4765–4774.
- [34] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: *Proceedings of the 34th International Conference on Machine Learning*, pp. 1321–1330.
- [35] Angelopoulos, A.N., Bates, S., 2021. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv:2107.07511*.
- [36] Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., et al., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40, 913–929.
- [37] Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., et al., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications* 11, 4540.
- [38] Olofsson, P., Foody, G.M., Herold, M., Stehman, S.V., Woodcock, C.E., Wulder, M.A., 2014. Good practices for estimating area and assessing accuracy of land change. *Remote Sensing of Environment* 148, 42–57.
- [39] Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollár, P., 2017. Focal loss for dense object detection, in: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2980–2988.
- [40] Devroye, L., Györfi, L., Lugosi, G., 1996. *A Probabilistic Theory of Pattern Recognition*. Springer, New York.
- [41] Shalev-Shwartz, S., Ben-David, S., 2014. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, Cambridge.
- [42] Breiman, L., Friedman, J.H., Olshen, R.A., Stone, C.J., 1984. *Classification and Regression Trees*. Wadsworth, Belmont, CA.

- [43] Goodfellow, I., Bengio, Y., Courville, A., 2016. Deep Learning. MIT Press, Cambridge, MA.
- [44] Vovk, V., Gammerman, A., Shafer, G., 2005. Algorithmic Learning in a Random World. Springer, New York.
- [45] Hoeffding, W., 1963. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association* 58, 13–30.
- [46] Hastie, T., Tibshirani, R., Friedman, J., 2009. The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd ed. Springer, New York.
- [47] Platt, J.C., 1999. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods, in: *Advances in Large Margin Classifiers*. MIT Press, pp. 61–74.